Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Aim:

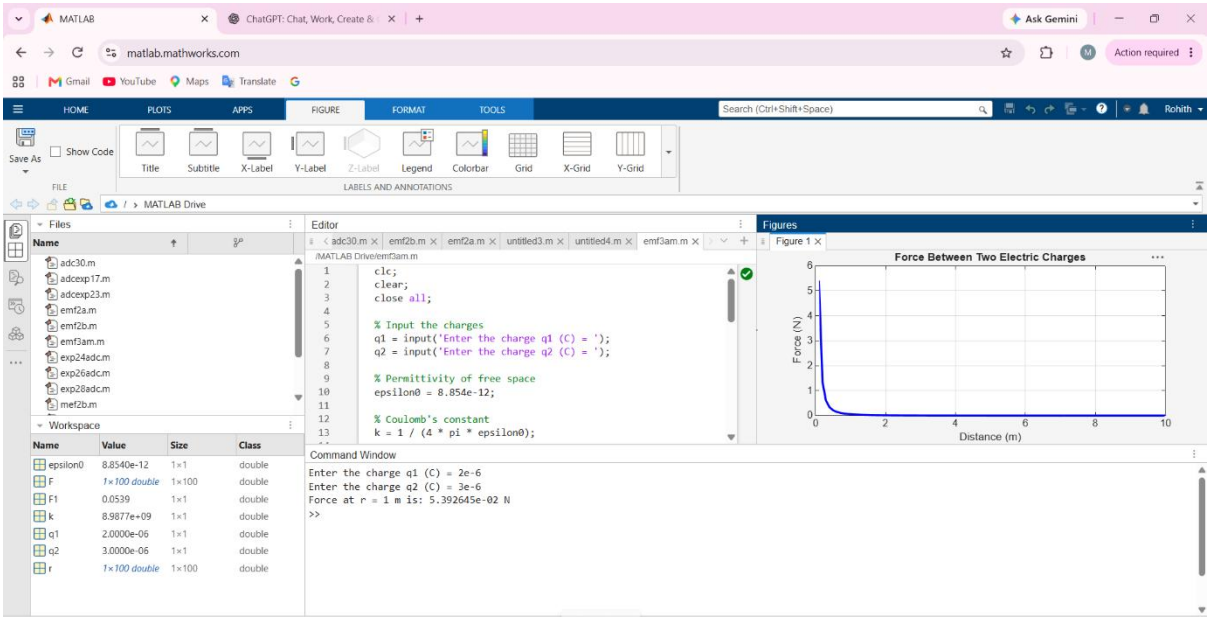
To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

Program:

Test case (i) Determination and plotting of force between two electric charges with varying distance

```
clc;
clear;
q1 = input('Enter the charge q1=');
q2 = input('Enter the charge q2=');
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
r = linspace(0.1,10,100);
F = k*q1*q2./(r.^2);
disp('Force at r = 1 m is:');
disp(k*q1*q2/(1^2));
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between two electric charges');
```

output



Test case (ii) Determination and plotting of force between two electric charges with different magnitudes

```
clc;

q1 = 5e-6;

q2 = 2e-6;

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.5,5,100);

F = k*q1*q2./(r.^2);

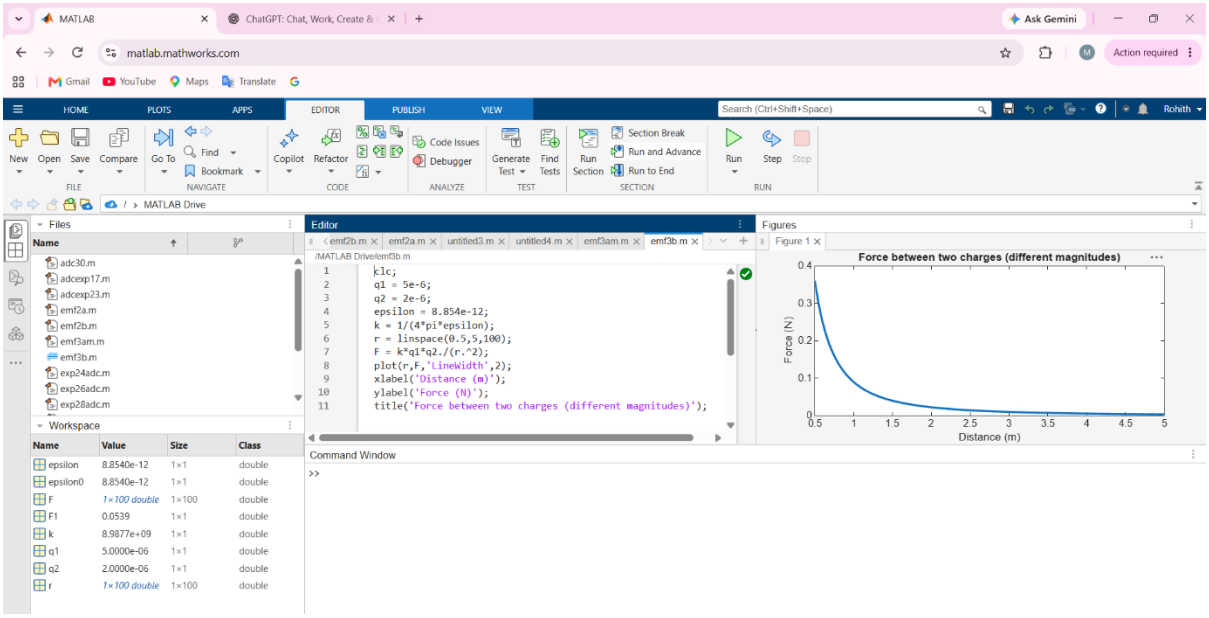
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two charges (different magnitudes)');
```

OUTPUT



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifie s most quantities and models correctl y.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25	2.5	
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel-current force calculations correctly with units.	Calculat ions are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	

3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissio ns.	Limited sweep or incomplete trend interpretatio n.	Does not establish a valid relationship.	30	3	
4	Explain and report results	Explain and report results	Explains attraction/rep ulsi on and reports assumptions, units and conclusions clearly.	Conclus ion is correct with minor docume ntation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20	2	
Tota l							100	10	